

AI-Powered UTI Antibiotic System

ANALYSIS REPORT • ACADEMIC RESEARCH DOCUMENT

Patient Demographics

Age / Gender	41.0 / Female
Department	Gynaecology
Diagnosis	Urethritis
UTI Classification	Recurrent UTI
Sample Type	Pus Swab

Clinical Presentation

Chief Complaints: Suprapubic pain; Burning urination

Comorbidities: None

Surgical History: None

Social History: Non smoker

Laboratory Results

Test Name	Value	Unit	Normal Range
Cbp Lymphocytes	38.0	%	20 - 40
Wbc	8950.0	cells/ μ L	4000 - 11000
Polymorphs	63.0	%	40 - 75
Crp	8.0	mg/L	< 10
Rft Serum Creatinine	0.7	mg/dL	0.6 - 1.3
Serum Uric Acid	4.0	mg/dL	3.5 - 7.2
Blood Urea	24.0	mg/dL	7 - 20
Cue Pus Cells	11.0	/HPF	0 - 5
Epithelial Cells	3.0	/HPF	0 - 5
Proteins	Trace		Negative
Rbc	1.0	/HPF	0 - 3

AI Pathogen Prediction

Predicted Organism	Enterococcus faecium
Bacteria Type	Gram-positive coccus (Higher risk of Vancomycin resistance - VRE)

Antibiotic Susceptibility Profile

Predicted Sensitive	Daptomycin
Predicted Resistant	Clarithromycin

Recommended Therapy

Daptomycin

Dosage: 6 mg/kg IV once daily (adjust to 6 mg/kg IV q24 h for serious Enterococcus faecium infections; weight not provided, use standard dosing for average adult).

Precautions: Renal function is normal (serum creatinine 0.7 mg/dL). No dose adjustment needed. Monitor creatine kinase (CK) for myopathy, especially with prolonged therapy. No known liver dysfunction; no known drug allergies reported.

Rationale: Daptomycin is a lipopeptide antibiotic that is active against vancomycin-resistant Enterococcus faecium. It is the recommended agent for VRE infections and is the only antibiotic on the predicted sensitive list for this patient.

Antibiotic Drug History

Clinical Summary

Patient Summary

- **Infection type & site:** Recurrent lower urinary tract infection (urethritis) confirmed by pus swab.
- **Organism identified:** Enterococcus faecium (Gram-positive coccus, high risk of vancomycin resistance).

Antibiotic profile

Resistance	Sensitivity
Clarithromycin (predicted resistant)	Daptomycin (predicted sensitive)

Recommended therapy

- **Daptomycin** 6 mg/kg IV once daily (adjust to 6 mg/kg IV q24 h for serious infections).
- *Rationale:* Daptomycin is the first-line agent for vancomycin-resistant *E. faecium* and is the only drug on the sensitive list.
- *Dosing note:* Weight not provided; use standard adult dosing.

Precautions & monitoring

- Renal function normal (serum creatinine 0.7 mg/dL); no dose adjustment required.
- Monitor serum creatine kinase (CK) for myopathy, especially with prolonged courses.
- No known liver dysfunction or drug allergies.
- Continue to reassess clinical response and repeat culture if symptoms persist.

This concise plan targets the identified pathogen with an evidence-based agent while accounting for the patient's renal status and potential adverse effects.